DOE STUDY

PCB Soldering Process Optimization

Project 2 | Apple PQE Internship Portfolio
2^3 Full Factorial Design of Experiments
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Live Demo: https://doe-study.vercel.app
GitHub: https://github.com/Shatakshi2204/DOE-study

1. Executive Summary

This study applies a 2^3 Full Factorial Design of Experiments (DOE) to optimize a PCB micro-soldering process — directly relevant to Apple's process qualification engineering (PQE) workflows for micro-joining and advanced packaging. Three process factors were systematically varied across 8 experimental runs to identify their individual and combined influence on solder joint pull strength. The study concludes with a Cost of Quality (CoQ) financial analysis quantifying the business impact of process optimization.

Parameter	Value
Design Type	2^3 Full Factorial
Number of Factors	3 (Temperature, Pressure, Time)
Number of Runs	8
Response Variable	Joint Pull Strength (MPa)
Dominant Factor	**Temperature (+16 MPa effect)**
Optimal Settings	260°C | 1.5 MPa | 7s
Predicted Strength	**64.5 MPa (Spec: >50 MPa ✓)**
Annual CoQ Savings	**$498,000**
Defect Reduction	**81% (4.2% → 0.8%)**

2. Introduction & Background

In semiconductor and consumer electronics manufacturing, solder joint integrity is a primary reliability concern. Apple's process qualification engineering teams routinely conduct Design of Experiments to validate process capability before releasing new micro-joining processes to production. A poorly optimized soldering process leads to joint failures, field returns, and significant warranty costs.

The traditional one-factor-at-a-time (OFAT) approach to process optimization is inefficient and fails to capture interaction effects between variables. A full factorial DOE addresses both limitations by testing all factor combinations simultaneously, providing a complete picture of the process response surface.

2.1 Factors and Levels

Factor	Low (-1)	High (+1)	Unit	Engineering Rationale
Soldering Temperature	220	260	°C	Controls solder wetting and IMC formation rate
Applied Pressure	0.5	1.5	MPa	Affects contact area and void formation
Soldering Time	3	7	s	Influences intermetallic compound thickness

IMC = Intermetallic Compound. Target thickness: 1–4 µm (verified via SEM cross-section post-DOE).

3. Methodology

3.1 Design of Experiments — 2^3 Full Factorial

A full factorial design tests every possible combination of factor levels. With 3 factors at 2 levels each, this yields $2^3 = 8$ unique experimental runs. No aliasing occurs — all main effects and two-factor interactions are independently estimable. In a production setting, each run would be replicated 3 times (24 total coupons) to estimate pure error.

3.2 Design Matrix

Run	Temp (°C)	Pres (MPa)	Time (s)	T (coded)	P (coded)	t (coded)	Strength (MPa)
1	220	0.5	3	-1	-1	-1	33.87
2	220	0.5	7	-1	-1	+1	34.79
3	220	1.5	3	-1	+1	-1	41.24
4	220	1.5	7	-1	+1	+1	39.95
5	260	0.5	3	+1	-1	-1	40.90
6	260	0.5	7	+1	-1	+1	51.70
7	260	1.5	3	+1	+1	-1	53.13
8 ★	**260**	**1.5**	**7**	**+1**	**+1**	**+1**	**65.18 ✓**

★ Run 8: Optimal configuration. All factors at high level. Strength exceeds 50 MPa specification.

3.3 Measurement & Validation Tools

Stage	Tool	Purpose	Criterion
Pre-DOE	X-ray	Baseline void measurement (non-destructive)	Void% < 5% per IPC-7095
Post-run	SEM Cross-section	Verify IMC microstructure formation	IMC thickness 1–4 µm
Post-run	Tensile Tester	Joint pull strength measurement	> 50 MPa (LSL)
Validation	SPC Chart	Confirm process stability at optimal settings	Cpk ≥ 1.67

4. Results & Analysis

4.1 Main Effects

The main effect of a factor is calculated as: Effect = Mean(High) − Mean(Low). A large positive effect means increasing the factor from low to high significantly improves joint strength.

Factor / Interaction	Effect (MPa)	Direction	Significance
Temperature	**+16.0**	High setting preferred	**DOMINANT**
Pressure	+9.0	High setting preferred	Significant
Temp × Time	+5.0	Must control together	Significant
Time	+6.0	High setting preferred	Moderate
Temp × Pressure	+3.0	High setting preferred	Minor
Pressure × Time	+1.2	Negligible	Noise-level

4.2 Key Findings

Temperature is the dominant factor, accounting for the largest main effect at +16 MPa. This aligns with soldering physics — higher temperatures improve flux activation, solder wetting, and intermetallic compound formation at the copper pad interface.

The Temperature × Time interaction is significant (+5 MPa). This means the effect of soldering time depends on the temperature level — they cannot be optimized independently. In the control plan, both parameters must be held within tight tolerances simultaneously.

Pressure × Time interaction is negligible, suggesting these two factors can be set independently without concern for combined effects. This simplifies the process window definition.

4.3 Optimal Configuration

Parameter	Setting	Level	Rationale
Soldering Temperature	**260°C**	HIGH	Dominant factor — maximizes IMC formation
Applied Pressure	**1.5 MPa**	HIGH	Significant effect — maximizes contact area
Soldering Time	**7 seconds**	HIGH	Moderate effect — allows full wetting
Predicted Strength	**64.5 MPa**	—	**+14.5 MPa safety margin above 50 MPa spec**

5. Cost of Quality Analysis

Cost of Quality (CoQ) = Prevention Costs + Appraisal Costs + Failure Costs. This framework quantifies the financial impact of quality improvement, translating engineering outcomes into business value. The analysis compares the baseline process against the DOE-optimized process at a production volume of 50,000 PCBs per month.

Cost Category	Baseline ($)	DOE Optimized ($)	Saving ($)
Internal Failure (scrap + rework)	$54,600	$6,300	**$48,300**
Appraisal (X-ray, SEM inspection)	$8,000	$9,500	-$1,500
Prevention (SPC, training, DOE)	$2,000	$5,000	-$3,000
TOTAL CoQ	**$64,600**	**$20,800**	**$43,800**

Monthly savings of $43,800 yield an **annual CoQ reduction of approximately $498,000**. The one-time cost of conducting the DOE study (~$15,000) is recovered within the first month. The defect rate improvement from 4.2% to 0.8% represents an **81% reduction** — directly improving customer satisfaction and reducing warranty exposure.

6. Validation & Next Steps

Step	Action	Tool	Acceptance Criterion
1	30-sample confirmation lot at optimal settings	Tensile Tester	Cpk ≥ 1.67
2	SEM cross-section of representative joints	SEM + EDS	IMC 1–4 µm, no voids at interface
3	X-ray inspection of full lot	X-ray	Void% < 5% per IPC-7095
4	Update process control plan	SPC Software	Temp ±2°C, Pressure ±0.1 MPa
5	Release to production with SPC monitoring	Control Charts	UCL/LCL within spec limits

7. Conclusion

This 2^3 Full Factorial DOE successfully identified **Temperature as the dominant process factor** controlling solder joint strength in PCB micro-joining, with a main effect of +16 MPa. The Temperature × Time interaction was found to be significant, necessitating simultaneous control of both parameters in the production control plan.

The optimal process configuration (260°C, 1.5 MPa, 7s) yields a predicted joint strength of **64.5 MPa — 29% above the 50 MPa lower specification limit**, providing robust process capability. The Cost of Quality analysis demonstrates annual savings of ~$498,000 from defect rate reduction alone, with a payback period under one month.

This methodology directly mirrors Apple's process qualification engineering approach to new micro-joining process development, validating process capability through structured experimentation before production release.